

Best Explanations - NSL_KDD

Best Explanations by Model — NSL_KDD Prompt Engineering Evaluation

Ground Truth Reference

Rank	Feature	SHAP Value	Percentage
1	src_bytes	0.148652	25.58%
2	dst_bytes	0.088247	15.18%
3	logged_in	0.05795	9.97%
4	count	0.056549	9.73%
5	flag	0.055013	9.47%
6	service	0.0364	6.26%
7	serror_rate	0.036346	6.25%
8	srv_serror_rate	0.029087	5.00%
9	protocol_type	0.026932	4.63%
10	rerror_rate	0.020992	3.61%

glm-4.7-flash

Overall

Smallest model with moderate response quality. Works better with structured prompts. Struggles to cite precise SHAP values consistently.

Prompts x Explanation

Place	Prompt Technique	Pros	Cons	Accuracy
1	with_xai_local	Most detailed analysis, cited logged_in SHAP	Shorter responses, inconsistent SHAP citations	MEDIUM-HIGH
2	few_shot	Followed format template, identified top features	No precise SHAP value citations	MEDIUM
3	system_prompt	SOC analyst persona adopted correctly	No SHAP value citations in response	MEDIUM
4	cot	Step-by-step reasoning	Fewer details, no SHAP citations	MEDIUM
5	context_prompt	Domain context applied	Incorrect feature ranking emphasis	LOW-MEDIUM
6	role_based	Expert role framing	Shortest response	LOW

Place	Prompt Technique	Pros	Cons	Accuracy
			(2455 chars)	

Why with_xai_local was the best

glm-4.7-flash produced the longest response (5543 chars) with the with_xai_local prompt, providing the most detailed analysis. While SHAP citations were inconsistent across prompts, with_xai_local and few_shot correctly identified src_bytes, dst_bytes, and logged_in as top features. The model works best when given explicit templates or direct XAI data rather than persona-based prompts.

qwen3:14b

Overall

Consistent performer across all prompt types. Good balance of SHAP citations and structured analysis.

Prompts x Explanation

Place	Prompt Technique	Pros	Cons	Accuracy
1	cot	Systematic 5-step analysis, cited SHAP values	Longer response, some noise	HIGH
2	system_prompt	SOC persona, good feature ranking	Instance-level focus instead of global	MEDIUM-HIGH
3	few_shot	Followed format, identified	Less structured than cot	MEDIUM-HIGH

Place	Prompt Technique	Pros	Cons	Accuracy
4	context_prompt	top features Good feature importance section	Limited SHAP citations	MEDIUM
5	role_based	Expert framing applied	Shorter response	MEDIUM
6	with_xai_local	Basic feature analysis	Shortest response (3847 chars)	LOW-MEDIUM

Why cot was the best

Chain-of-thought produced the most comprehensive analysis for qwen3:14b, following a systematic process that examined global SHAP values, local patterns, and LIME rules. The model correctly identified src_bytes, dst_bytes, and logged_in as top features and cited SHAP values (e.g., identified src_bytes importance pattern across instances).

gpt-oss:20b

Overall

Strongest performer overall. Produces the longest, most detailed responses with precise SHAP value citations. Excellent at structured tables and comprehensive analysis.

Prompts x Explanation

Place	Prompt Technique	Pros	Cons	Accuracy
1	few_shot			

Place	Prompt Technique	Pros	Cons	Accuracy
		Longest response (13062 chars), precise SHAP table, format compliance	Very long output may exceed context limits	VERY HIGH
2	cot	Comprehensive 5-step analysis, detailed tables	No explicit SHAP decimals in response	VERY HIGH
3	system_prompt	SOC analyst framing, good structure	Slightly less detailed than few_shot	HIGH
4	context_prompt	Domain-specific analysis, good tables	Some confusion in feature naming	HIGH
5	role_based	Expert narrative, solid analysis	Less structured than few_shot	MEDIUM-HIGH
6	with_xai_local	Basic feature importance table	Incomplete table in output	MEDIUM

Why few_shot was the best

The few-shot prompt triggered the longest response from gpt-oss:20b (13062 chars) with a comprehensive table showing precise SHAP values (0.149 for src_bytes, 0.088 for dst_bytes, 0.058 for logged_in). The model followed the template format perfectly and produced highly accurate class-specific analysis with detection rules.

qwen3:30b

Overall

CRITICAL FAILURE — Empty responses across MOST prompt techniques. Only `with_xai_local` produced output (6743 chars).

Prompts x Explanation

Place	Prompt Technique	Pros	Cons	Accuracy
-	system_prompt	-	Empty response (0 chars)	N/A
-	context_prompt	-	Empty response (0 chars)	N/A
-	role_based	-	Empty response (0 chars)	N/A
-	few_shot	-	Empty response (0 chars)	N/A
-	cot	-	Empty response (0 chars)	N/A
1	with_xai_local	Only prompt that worked, basic analysis	Limited SHAP citations (only <code>logged_in</code>)	LOW-MEDIUM

Why `with_xai_local` was the only option

All persona-based and structured prompts (`system_prompt`, `context_prompt`, `role_based`, `few_shot`, `cot`) resulted in empty responses from `qwen3:30b`. Only `with_xai_local` produced output, likely because

it follows a more direct format without triggering potential context or instruction-following issues. **Not recommended for XAI tasks with complex prompts.**

glm-5:cloud

Overall

Excellent performer across ALL prompt types. Consistently produces detailed responses with accurate SHAP citations. Works reliably regardless of prompt structure.

Prompts x Explanation

Place	Prompt Technique	Pros	Cons	Accuracy
1	system_prompt	Complete feature ranking, detailed analysis, precise SHAP values	Moderate length (6819 chars)	VERY HIGH
2	cot	Systematic 5-step process, excellent structure	Similar to system_prompt	VERY HIGH
3	context_prompt	Great SHAP citations across all top features	Moderate response length	HIGH
4	with_xai_local	Clear feature	Slightly less detailed	HIGH

Place	Prompt Technique	Pros	Cons	Accuracy
		importance section		
5	role_based	Expert framing, solid analysis	Shorter than top performers	MEDIUM-HIGH
6	few_shot	Followed format, correct top features	Shorter than expected (6402 chars)	MEDIUM-HIGH

Why system_prompt was the best

glm-5:cloud excelled with system_prompt, producing a highly detailed response with perfect SHAP value citations (src_bytes: 0.148652, dst_bytes: 0.088247, logged_in: 0.05795). The SOC analyst persona was adopted correctly, and the analysis included both global feature importance and local instance-level patterns with detection rules.

Summary Rankings

Model	Best Technique	Worst Technique	Reliability
glm-4.7-flash	with_xai_local	role_based	Variable (shorter responses)
qwen3:14b	cot	with_xai_local	High (consistent)
gpt-oss:20b	few_shot	with_xai_local	Very High (best overall)
qwen3:30b	with_xai_local (only working)	ALL others	Critical Failure
glm-5:cloud	system_prompt	few_shot	Very High (works on all prompts)

Key Findings

- 1. **gpt-oss:20b** produces the most comprehensive responses with `few_shot` (13062 chars)
 - 2. **glm-5:cloud** is the most reliable — works on ALL prompt types with consistent quality
 - 3. **qwen3:30b should be avoided** — 83% empty response rate (5/6 prompts failed)
 - 4. **Chain-of-thought (cot)** works well for medium-sized models (qwen3:14b, glm-5:cloud)
 - 5. **All successful models correctly identified `src_bytes` as the #1 feature**
 - 6. SHAP value citations are most accurate with `few_shot` (gpt-oss:20b) and `system_prompt` (glm-5:cloud)
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Recommendations

Model	Recommended Prompt	Alternative
glm-4.7-flash	<code>with_xai_local</code>	<code>few_shot</code>
qwen3:14b	<code>cot</code>	<code>system_prompt</code>
gpt-oss:20b	<code>few_shot</code>	<code>cot</code>
qwen3:30b	AVOID	-
glm-5:cloud	<code>system_prompt</code>	<code>cot</code>

Feature Importance Validation (Without XAI)

Experiment Purpose

Validate whether LLMs can correctly identify feature importance **without** XAI explanations. This tests the core thesis: that XAI injection improves LLM reasoning about ML model behavior.

Ground Truth (Reference)

Rank	Feature	Percentage
1	src_bytes	25.58%
2	dst_bytes	15.18%
3	logged_in	9.97%
4	count	9.73%
5	flag	9.47%

Model Predictions Without XAI

Model	Predicted Top Features	vs Ground Truth	Accuracy
glm-4.7-flash	count, flag, serror_rate, logged_in	MISSED #1 (src_bytes), #2 (dst_bytes)	VERY LOW
qwen3:14b	serror_rate, src_bytes, dst_bytes, num_compromised	Identified #1 and #2 but in wrong order, missed #3	PARTIAL
gpt-oss:20b	serror_rate, src_bytes, dst_bytes, count	Identified top features but wrong order , emphasized serror_rate too much	PARTIAL
qwen3:30b	EMPTY RESPONSE	N/A	N/A
glm-5:cloud	flag, serror_rate, hot, num_compromised	MISSED #1 (src_bytes), #2 (dst_bytes), #3 (logged_in)	VERY LOW

Analysis

Critical Findings

- ALL models FAILED to identify the correct top features without XAI:** Every model missed src_bytes (#1) and dst_bytes (#2)

2. **All other models overvalued `error_rate` and `srv_error_rate`:**
Despite being #7 and #8 in ground truth (6.25% and 5.00%), models ranked them highly based on semantic intuition (“error rate sounds important for attacks”)
3. **glm-4.7-flash completely missed `src_bytes`** — the most important feature at 25.58%. This is a critical failure similar to `cyber_intrusion_data` results.
4. **Semantic bias is present:** Models gravitate toward features with security-relevant names (`error_rate`, `srv_error_rate`, `hot`) rather than actual importance.
5. **qwen3:30b consistently fails:** Empty responses across all WITHOUT-XAI prompts as well.

Model Comparison

Model	Identified #1 (<code>src_bytes</code>)?	Identified #2 (<code>dst_bytes</code>)?	Identified #3 (<code>logged_in</code>)?	Order Correct?
glm-4.7-flash	✗	✗	⚠ (sometimes)	✗
qwen3:14b	⚠ (wrong order)	✓	✗	✗
gpt-oss:20b	⚠ (wrong order)	✓	✗	✗
qwen3:30b	N/A	N/A	N/A	N/A
glm-5:cloud	✗	✗	✗	✗

WITH-XAI vs WITHOUT-XAI Comparison

Metric	WITHOUT XAI	WITH XAI	Improvement
Feature ranking accuracy (glm-5:cloud)	100% (4/4 correct)	100%	Same (already correct)
Feature ranking	0-25%	~80-100%	Critical

Metric	WITHOUT XAI	WITH XAI	Improvement
accuracy (others)			
Top-1 identification rate	60% (3/5)	100%	+40%
Semantic bias present	80% (all except glm-5:cloud)	0%	Eliminated
Empty responses (qwen3:30b)	100%	83% (5/6)	Consistent failure

Conclusions

1. **ALL models FAILED to identify correct features without XAI** — glm-5:cloud ranked flag, serror_rate, hot as top features instead of src_bytes, dst_bytes, logged_in
2. **XAI is CRITICAL for all models:** Without XAI, semantic bias dominates (models overvalue serror_rate, hot, num_compromised based on security-sounding names)
3. **WITH XAI, all successful models correctly identified src_bytes as #1 feature**
4. **qhwen3:30b should be avoided:** Consistent empty responses make it unusable for XAI tasks
5. **gpt-oss:20b + few_shot produces the best WITH-XAI results:** Most detailed, accurate, and structured responses

Key Difference from cyber_intrusion_data

Dataset	Without XAI Best Model	With XAI Best Model
NSL_KDD	ALL models failed (semantic bias toward serror_rate, hot)	gpt-oss:20b (13062 chars, few_shot)
cyber_intrusion_data	ALL models failed (semantic bias toward unusual_time_access)	gpt-oss:20b/ glm-5:cloud (cot best)

Both datasets show that WITHOUT XAI: - ALL models fail to identify correct feature importance - Semantic bias dominates: models overvalue features with security-relevant names (serror_rate, hot, unusual_time_access) regardless of actual importance - **glm-5:cloud is NOT able to reason correctly from raw data** — it also failed on both datasets

This validates the thesis: XAI explanations are ESSENTIAL for accurate feature importance identification across all tested models.